

## Project Design Phase-II

### Solution Requirements (Functional & Non-functional)

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 4 September 2026                             |
| Team ID       | SWUID2026191970                              |
| Project Name  | Metro Ticket Generating System in ServiceNow |
| Maximum Marks | 4 Marks                                      |

#### Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                        |
|--------|-------------------------------|---------------------------------------------------------------------------|
| FR-1   | Station Selection             | Select Starting From station<br>Select Going To station                   |
| FR-2   | Journey Selection             | Select Single Journey or Return Journey                                   |
| FR-3   | Passenger Selection           | Select number of passengers for the ticket                                |
| FR-4   | Fare Calculation              | Automatically calculate fare based on journey and passenger details       |
| FR-5   | Payment Selection             | Select payment mode: UPI, Card, or Others                                 |
| FR-6   | Ticket Booking                | Submit metro ticket booking request through the ServiceNow Catalog Item   |
| FR-7   | QR Ticket Generation          | Generate and display a QR code for the metro ticket                       |
| FR-8   | Station Details Management    | Store and maintain metro station details in the Station Details table     |
| FR-9   | Ticket Record Management      | Create and maintain ticket/request information in ServiceNow              |
| FR-10  | Access Control                | Allow authorized users to perform permitted operations on project records |

#### Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| NFR No. | Non-Functional Requirement | Description                                                                                                                                                                          |
|---------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | Usability                  | The ticket booking form should be simple and easy to use, allowing passengers to select stations, journey type, passengers, payment mode, and submit the request with minimal steps. |
| NFR-2   | Security                   | ServiceNow access controls should protect station and ticket-related records and restrict operations to authorized users.                                                            |
| NFR-3   | Reliability                | The system should consistently calculate fares, process ticket requests, and display the generated QR ticket without unexpected failures.                                            |
| NFR-4   | Performance                | The booking form, fare calculation, and QR display should respond quickly to                                                                                                         |

|       |              |                                                                                                                                                                       |
|-------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |              | provide a smooth ticket-booking experience.                                                                                                                           |
| NFR-5 | Availability | The solution should be available through the ServiceNow platform whenever the ServiceNow instance and required services are operational.                              |
| NFR-6 | Scalability  | The solution should support increasing numbers of stations, passengers, ticket requests, and transaction records without requiring major changes to the basic design. |